

Related-Work

Study	Core Method & Architecture	Dataset & Target Focus	Key Findings & Results	Limitations & Drawbacks
SE-DenseNet Network (2025)	SE-DenseNet: Integrates Squeeze-and-Excitation (SE) attention between DenseBlock & Transition layers.	Ninapro DB2 & DB4 (12 electrode channels, 49–52 gestures).	Achieved 85.93% (DB2) and 82.39% (DB4) gesture recognition accuracy.	High parameter footprint; prone to gradient vanishing on deeper inaccurate layers.
Spatiotemporal GCN (2024)	ST-GCN: Models physical forearm electrode geometry as graph nodes for spatial aggregation.	Public multi-channel EMG benchmarks (8–12 channels).	Outperformed classical CNNs; maintains high accuracy under reduced sensor nodes.	Assumes fixed electrode graph connections; sensitive to electrode displacement.
Hybrid CNN-ViT (2025)	Convolutional Vision Transformer (CViT): Hybrid local convolutions + global self-attention.	Ninapro DB3 (Amputees) & Compact Myo (8 channels).	Achieved 94.2% on Ninapro DB3 amputee subjects (vs. 92.0% CNN baseline).	Transformer self-attention overhead increases computational latency on microcontrollers.

Real-Time Graph Model (2026)	sEMG Graph Neural Network: Direct muscle activation graph construction + fast GNN.	8-channel sEMG armband array (8 subjects).	Reached 99% accuracy with a low inference budget of 48 ms execution time.	Tested primarily on intact, homogeneous subjects; lacks validation on amputee DBs.
Multistream Hybrid TCN-Transformer (2026)	TCN + Bi-LSTM/GRU + Transformer + ArcFace: Combines temporal dynamics & angular margin loss.	FORS-EMG dataset (Multi-gesture surface EMG).	Reached 96.4% within-subject accuracy using feature-space margin constraints.	Accuracy dropped to 74% in cross-subject settings, revealing severe domain shift.

Research Gaps of these paper:

Paper: *Spatiotemporal Graph Neural Network* (Lee et al., 2024)

Research Gap: While Lee et al. introduced graph structures to capture muscle spatial proximity, their model relies on static physical sensor topologies. It fails to dynamically reconfigure graph edges when inter-channel muscle synchronization changes during complex gesture transitions or electrode shifts.

Papers: *SE-DenseNet Network* (2025) and *Hybrid Convolutional Vision Transformer* (2025)

Research Gap: Both SE-DenseNet (2025) and CViT (2025) achieve high classification accuracy on Ninapro datasets by stacking complex attention blocks (Squeeze-and-Excitation / Transformer self-attention). However, their large parameter footprints introduce computational

latency that exceeds the strict <300 ms end-to-end execution budget required for real-time robotic controllers and myoelectric prostheses.

Paper: *Multistream Hybrid TCN-Transformer* (2026)

Research Gap: The hybrid TCN-Transformer model achieves about **96.4% accuracy** when it is trained and tested on the **same person (intra-subject)**. However, its accuracy drops to about **74%** when it is tested on **different people (cross-subject)**. This happens because people have different muscle signal patterns, different amounts of fat under the skin that affect electrical signals, and slight differences in where the electrodes are placed. These variations make it difficult for the model to perform consistently across different users.

Paper: *Fast sEMG Graph Neural Network* (2026)

Research Gap: The fast sEMG-GNN model (2026) achieves an impressive 48 ms inference time with 99% accuracy, but it was evaluated exclusively on intact, homogeneous subjects using basic 8-channel armbands. It lacks validation on transradial amputee datasets (e.g., **Ninapro DB3**), where signal SNR is significantly lower due to tissue atrophy and remnant muscle distortion.

Papers: *STDC-Net* (2026) & General Baseline Benchmarks

Identified Research Gap: Recent studies (e.g., STDC-Net) have begun integrating SHAP-based feature importance, but the vast majority of recent sEMG literature still relies solely on simple point-estimate mean accuracy without conducting formal non-parametric statistical significance testing (such as Dietterich's 5 times 2v paired t-test or Demsar's Friedman-Nemenyi test) across multi-subject datasets.

Reference

1. **SE-DenseNet (2025):** *Gesture recognition from surface electromyography signals based on the SE-DenseNet network*, ResearchGate / Ninapro DB2 & DB4 Benchmark.

https://www.researchgate.net/publication/388487726_Gesture_recognition_from_surface_electromyography_signals_based_on_the_SE-DenseNet_network

2. **ST-GCN (2024):** *Decoding Gestures in Electromyography: Spatiotemporal Graph Neural Networks for Generalizable and Interpretable Classification*, IEEE TNSRE, 2024.

<https://ieeexplore.ieee.org/document/10818442/>

3. **CViT (2025):** *Hybrid Convolutional Vision Transformer for Robust Low-Channel sEMG Hand Gesture Recognition: A Comparative Study with CNNs*, MDPI Sensors, 2025.

<https://www.mdpi.com/2313-7673/10/12/806>

4. **Fast sEMG-GNN (2026):** *A Graph Neural Network Model for Real-Time Gesture Recognition Based on sEMG Signals*, arXiv:2607.07850, 2026.

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5. **Multistream TCN-ArcFace (2026):** *Surface EMG-Based Hand Gesture Recognition Using a Hybrid Multistream Deep Learning Architecture*, MDPI Sensors, 2026.

https://www.researchgate.net/publication/380646797_Unveiling_EMG_semantics_a_prototype-learning_approach_to_generalizable_gesture_classification